Pressure

Pressure is the push on a surface created by one or more forces. How much pressure is exerted depends upon the strength of the forces and the area of the surface. Walk over snow in showshoes and you won't sink in—but walk on grass in stiletto heels and you will.

Solids, liquids, and gases can apply pressure onto a surface because of their weight pressing down on it. The pressure applied by liquids and gases can be increased by squashing them. Pressure is measured in pounds per square inch (psi) or in newtons per square meter (N/m^2)—also called Pascals (Pa).

Atmospheric and water pressure

Near sea level, the weight of the air around us presses with a force of about 15 psi (100,000 Pa). Pressure decreases with altitude, because there is less air above pressing down. In the ocean, pressure increases quickly with depth, since water is denser than air.

ABOVE SEA LEVEL

130,000 ft (40,000 m)

115,000 ft (35,000 m)

98,000 ft (30,000 m)

82,000 ft (25,000 m)

65,500 ft (20,000 m)

49,000 ft (15,000 m)

130,000 ft (40,000 m)

Felix Baumgartner makes a record-breaking skydive from 127,852 ft (38,964 m).

Pilots of fighter jets cruising at 50,000 ft (15,000 m) wear pressure suits.

250 miles (400 km)
As a Soyuz spacecraft travels to the International Space Station (ISS), which orbits at 250 miles (400 km), gas molecules are so few and far between that air pressure is almost nonexistent. The space station's atmosphere is maintained at the same pressure as sea level.

115,000 ft (35,000 m)
As weather balloons ascend into the stratosphere, they expand from 6 ft 6 in (2 m) to 26 ft (8 m) across as air pressure decreases to just 0.1 psi (1,000 Pa). The gas molecules within the balloon spread out as pressure from outside diminishes.

60,000 ft (18,000 m)
Above this altitude—the Armstrong limit—humans cannot survive in an unpressurized environment. Air pressure is 1 psi (7,000 Pa) and exposed body fluids such as saliva and moisture in the lungs will boil away—but not blood in the circulatory system.

36,000 ft (111,000 m)
This is the cruising altitude of passenger jets. As a plane lifts off, your ears may pop due to the change in pressure: air trapped in the inner ear stays at the same pressure, but air pressure outside changes, exerting a force on your eardrum. Pressure falls to 3 psi (23,000 Pa) on the plane's exterior.

28,871 ft (8,848 m)
At Everest's summit, atmospheric pressure is one third of that at sea level: 4.5 psi (33,000 Pa). It is hard to make tea as water boils at 162ºF (72ºC)—not hot enough for a good brew. Liquids boil when the particles of which they are made move fast enough to have the same pressure as air—so when pressure falls, the boiling point is lower.